Sahaj Bhadja

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Summary

Software engineer with **2+ years of experience** building **secure backend APIs**, **SQL-backed services**, and **healthcare-adjacent platforms** across **Python, Flask, MySQL/PostgreSQL, AWS**, and cloud-native workflows for sensitive production systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Architected a **multi-tenant platform** on Azure SQL and Blob Storage with row-level security for **500+ users**, reducing dashboard latency from **1.2s to 350ms** through schema design, indexing, and query optimization.
- Built **FastAPI** services across PostgreSQL, SQL Server, and cloud storage, processing **5K+ monthly jobs** with validation, debugging, and release checks that sustained **99%+ production reliability**.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Developed and tested a **Flask + React** grading platform serving **150+ instructors**, handling **10K+ submissions** with secure ingestion, scalable APIs, and production-ready backend workflows.
- Designed concurrent services on **AWS RDS** with REST APIs and **GitHub Actions CI/CD**, sustaining **10K+ requests/day** while reducing turnaround time by **60%**.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform microservice software with **Jenkins**, PyTest, and Mockito, automating build, test, and release flows across **3 operating systems** with **20+ unit/integration suites**.
- Optimized API and ETL workflows using **AWS Glue**, **S3**, and OOP patterns, improving response time by **30%**, boosting pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a **MERN** resource-management platform with **REST APIs**, WebSockets, and microservices, supporting **8+ workflows**, structured account permissions, and **2M+ synchronized records**.
- Modeled users, resources, invoices, payments, and notification records in **Azure SQL** while using Jenkins-driven testing to improve release reliability and reduce resource wastage by **42%**.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built a cross-platform booking platform with **Flask**, React, TypeScript, and Firebase, supporting **50K+ booking records** with sub-**200ms** APIs and real-time mobile/web synchronization.
- Integrated webhook-driven payment, refund, and status-update flows for sensitive transactions, improving booking confirmation speed by **40%** and reducing manual refund handling by **70%**.

Skills

Languages: Python, PHP, Java, C++, JavaScript/TypeScript, SQL
Backend & APIs: Flask, FastAPI, Laravel, Node.js, Express.js, REST APIs, Microservices
Frontend: React.js, Next.js, TailwindCSS
Databases: MySQL, PostgreSQL, MongoDB, SQL Server, Azure SQL
Cloud & Infrastructure: AWS (EC2, EKS, RDS, S3, CloudFormation), Microsoft Azure, Docker, Kubernetes, GitHub Actions, Jenkins
Healthcare & Security: HIPAA-aware workflows, Secure API Design, Role-Based Access Control, Sensitive Data Handling, Observability
Engineering Practices: CI/CD, Unit & Integration Testing, Query Optimization, Indexing Strategies, Distributed Systems, Technical Documentation, Agile

Projects

HMIS Live | *MERN, TypeScript, Twilio, Appwrite*
- Built a healthcare platform supporting **100+ daily appointment workflows**, integrating ABDM APIs, secure patient access, and monitored backend operations that reduced record retrieval time by **2x**.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*
- Built a secure fintech platform supporting **50K+ users** and **1M+ transactions** with tokenized APIs, containerized deployment, and cloud-native data access controls.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, Clerk*
- Built a secure document platform over **50K+ documents** with authenticated access, semantic retrieval, and collaborative workflows that reduced document review time by **60%**.